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Toward Integrated AGI Ecosystems: A Review, Framework, and Research Agenda for Cyber-Physical-Social-Thinking Intelligence

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Abstract—Artificial General Intelligence (AGI) represents a transformative frontier of machine intelligence, aiming for human-level adaptability across diverse domains. However, AGI research remains fragmented, with most advancements focusing narrowly on algorithmic, computational, or neuro-inspired components. This review introduces Cyber-Physical-Social-Thinking (CPST) perspective to understand how AGI orchestrates complex multi-domain interactions, offering a unifying framework as an integrated intelligence system. Following PRISMA 2020 guidelines, we analyzed and 139 peer-reviewed studies (2021–October 2025), identifying three dominant research trajectories: (1) cognitive architecture integration, (2) social intelligence in distributed environments, and (3) reality-aware adaptation mechanisms. Our analysis identifies critical challenges in ethical governance, system resilience, and scalable learning across heterogeneous domains. We propose a forward-looking research agenda emphasizing hybrid human–machine intelligence, trustworthy AGI systems, and sustainable CPST ecosystems. This study advances intelligent systems research by providing an integrative framework that bridges AI algorithms with real-world deployment considerations, human factors, and societal impacts essential dimensions for next-generation intelligent technologies.

Index Terms—Artificial General Intelligence, Cyber-Physical-Social-Thinking Systems, Cognitive Orchestration, Human-AI Collaboration, Social Intelligence.

I. INTRODUCTION

THE concept of Artificial General Intelligence (AGI), has transitioned from theoretical concepts to a practical area of intense research due to the incessant human activity that today connect all the answers in such diverse areas as large-scale foundational models, multimodal learning, embodied robotics, complex simulation environments, and neuro-

symbolic reasoning. What is more, the merging of Cyber-Physical systems (CPS) with social computing and artificial intelligence has spurred the establishment of Cyber-Physical-Social-Thinking (CPST) systems, a complex network that embodies computational intelligence, physical infrastructure, human social dynamics, and cognitive processes [1], [2], [3], [4], [5], [6], [7], [8]. This notion means a radical transformation from specific automation to comprehensive intelligent environments that are capable of navigating multifarious interactions across digital, physical, social, and cognitive domains all at once [9], [10], [11], [12], [13], as represented on Fig. 1.

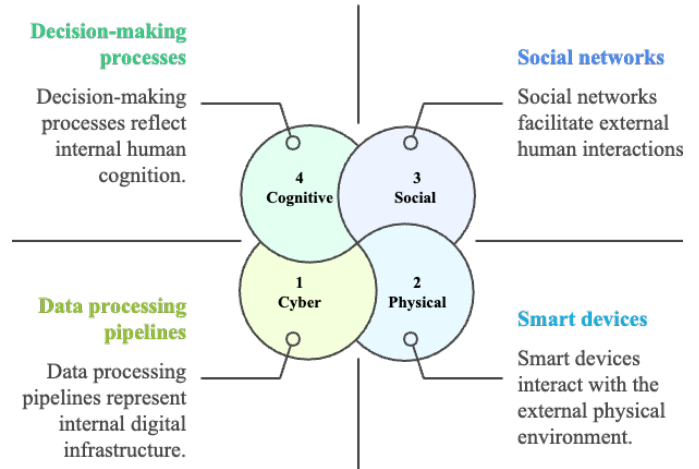


Fig. 1. The Four CPST Domains: Definition and Hierarchical Relationship

Meanwhile, the internet of Things (IoT) technologies serve as a facilitating factor that enhances the connection beyond limits [14], [15], and cognitive computing breakthroughs provide enhanced understanding of human reasoning [16], [17], [18]. Contrarily, the current Artificial Narrow Intelligence (ANI) systems such as the ones in Cyber-Physical-Systems are neither interconnected nor promoted with cognitive computing functionalities. In general, they are system designs oriented towards one task and therefore, they cannot cover the whole range of human intelligence, yet they are effective for instance in voice recognition [19] or energy optimization [20], [21] but they need the inter-domain reasoning skills that are not complementary to CPST Orchestration [22]. The existing disparity is manifested as separation in operations that do not support and solve the challenges across the parallel CPST dimensions, resulting stakeholder conditions [20], [21]. AGI is the solution.

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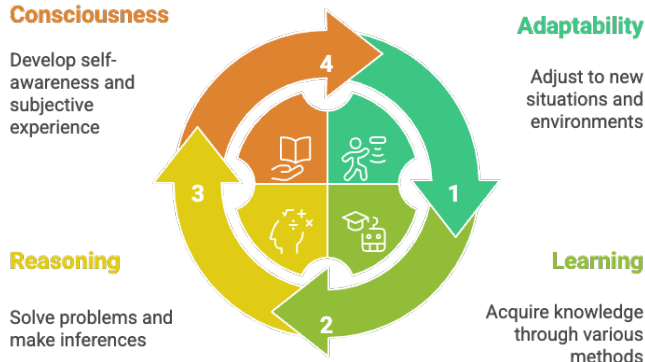


Fig. 2. Core principles for an Artificial General Intelligence (AGI) system.

Tracing the results from research [23], [22], [10], [24], it was known that AGI is comprised of four core principles such as Adaptability, Learning, Reasoning, and another optional but debated [25], [26], Consciousness that served as a building blocks for its design and operation. These principles are not final but point out that they the crucial parts of intelligence that an AGI system is supposed to have, as illustrated in Fig. 2. AGI facilitates reasoning across all CPST domains, supports adaptiveness in learning by meta-learning, which transfers the knowledge, and promotes a deep encapsulation of cultural, physical, and cognitive aspects[27], [28], [29]. The CPST framework, with its established capabilities, reaches further than the previous paradigms simply by pointing out these cyber, physical, social, and thinking dimensions of being the primary factors[30], [24], [31], [32]. However, the technicality of translating the AGI capabilities to the effective orchestration that would dynamically direct the complex interaction in four domains at a time still remain challenge.

A. CPST Framework and AGI orchestration Gaps

Fundamentally, this review employs CPST perspective to understand *how AGI orchestrates complex multi-domain interactions*, addressing the critical challenge of coordinating intelligence across heterogeneous domains that characterize modern computational ecosystems. The CPST framework represents a fundamental shift from domain-specific automation to holistic intelligent environments[1], [33]. However, as summarized in Table I, current implementations face critical limitations in four key areas that prevent effective orchestration [20], [34], [21], [35]. AGI, when analyzed through the CPST perspective, addresses these gaps through its unique capabilities for cross-domain reasoning and contextual adaptation [22], [44]. However, significant challenges remain in applying the CPST perspective to AGI capabilities for effective Orchestration patterns identified through CPST lens that can bridge these identified gaps[22], [23]. These challenges are particularly acute in applications requiring sophisticated human-AI collaboration across healthcare [45], [39], [46], industrial [47], [48], and urban domains [49].

TABLE I
CRITICAL GAPS IN CURRENT CPST ORCHESTRATION APPROACHES

Gap Category	Current Limitations	Impact on CPST Systems	Ref.
Architectural Fragmentation	Siloed domain-specific solutions; Lack of cross-domain coordination mechanisms	Inefficient resource allocation; Limited system-wide optimization	[36], [37], [29]
Social-Cognitive Integration	Basic intention recognition only; Limited theory of mind capabilities	Poor human-AI collaboration; Inability to understand complex social contexts	[34], [38], [39]
Evaluation Methodologies	Domain-specific metrics only; No holistic CPST performance measures	Inability to assess cross-domain orchestration effectiveness	[40], [4]
Ethical Governance	No frameworks for cognitive privacy; Limited accountability mechanisms	Risks to mental autonomy; Unclear responsibility for system decisions	[41], [42], [43]

B. Research Motivation

The orchestration of CPST systems represents a fundamental challenge in intelligent systems, requiring coordination across heterogeneous cyber, physical, social, and cognitive dimensions [1]. While AGI offers promising orchestration capabilities, the research landscape remains fragmented with critical methodological gaps. Current evaluations overwhelmingly favor technical metrics while neglecting socio-cognitive dimensions and ethical considerations [10], exacerbated by absent standardized benchmarks and limited multi-modal datasets. This systematic review addresses these dual challenges by establishing architectural foundations while identifying critical measurement gaps impeding robust CPST development.

C. Research Questions and Contributions

To systematically investigate these challenges, this review addresses five research questions designed to examine AGI orchestration capabilities in CPST perspectives:

- **RQ1 (Architectural Foundations):** What architectural paradigms enable effective AGI orchestration capabilities across CPST domains?
- **RQ2 (Cognitive Capabilities):** What cognitive capabilities are essential for AGI orchestrators in multi-domain environments?
- **RQ3 (Adaptation Mechanisms):** What adaptation mechanisms support performance in dynamic CPST contexts?
- **RQ4 (Socio-Technical Challenges):** What technical, ethical, and governance challenges emerge in AGI-mediated orchestration?
- **RQ5 (Evaluation Methodologies):** What evaluation methodologies effectively assess AGI performance across CPST domains?

This review makes five concrete contributions: (1) a comparative taxonomy of AGI orchestration capabilities architectures with performance trade-off analysis across reviewed studies; (2) a layered cognitive capability framework identifying critical gaps in social intelligence and theory-of-mind implementations; (3) a systematic analysis of adaptation mechanisms;

(4) an evidence-based challenge mapping demonstrating the accountability gap in distributed multi-agent systems; (5) a comprehensive evaluation methodology analysis exposing critical metric imbalances.

D. Paper Organization

The remainder of this review is organized as follows: Section II introduces the Background and theoretical foundations of CPST systems and AGI architectures. Section III outlines the methodology used in this study. Section IV synthesizes the literature through a taxonomic classification of architectural patterns and cognitive capabilities. Section V reports the systematic analysis of findings and synthesis. Section VI consolidates the identified challenges, Section VII proposes a research agenda, while section VIII concludes with theoretical and practical implications. Supporting materials are provided in the appendices to ensure transparency and reproducibility.

II. BACKGROUND AND THEORETICAL FOUNDATIONS

This section establishes the background and theoretical foundations necessary for understanding AGI orchestration capabilities in CPST systems, a critical intersection for intelligent system research. The CPST framework, illustrated in Fig. 3, provides the analytical lens for examining how AGI coordinates complex multi-domain interactions, offering a structured approach to examine coordination mechanisms across all four dimensions simultaneously addressing the core focus on integrated intelligent systems. We trace the parallel evolution of intelligent systems architectures and artificial intelligence capabilities, analyze the CPST multidimensional architecture, and examine AGI's role as a cognitive orchestrator across diverse application domains.

A. Co-Evolution of Intelligent Systems and Artificial Intelligence

The emergence of CPST as a viable paradigm is intrinsically linked to the parallel evolution of Artificial Intelligence. As intelligent systems grew in complexity from CPS to CPST (CPS → CPSS → CPST), the required form of intelligence to manage them evolved correspondingly from Automation to AGI (Automation → Narrow AI → AGI). Fig. 4 illustrates this symbiotic relationship, demonstrating how advancements in AI capabilities have consistently enabled transitions to more complex system paradigms[5], [29]. The development of CPST systems represents an evolutionary trajectory from earlier paradigms, as summarized in Table II. This evolution shows the increasing complexity of intelligent systems and the corresponding need for more sophisticated AI capabilities, culminating in the requirement for AGI orchestration capabilities in CPST environments.

1) *From Automation to Narrow AI in CPS/CPSS*: Early CPS relied on embedded software and basic automation for control and monitoring [50], [53], [59], [60]. These systems operated with pre-defined rules and required constant human oversight for any deviation from expected scenarios. The integration of sensing and actuation capabilities [61], [62], [63] created

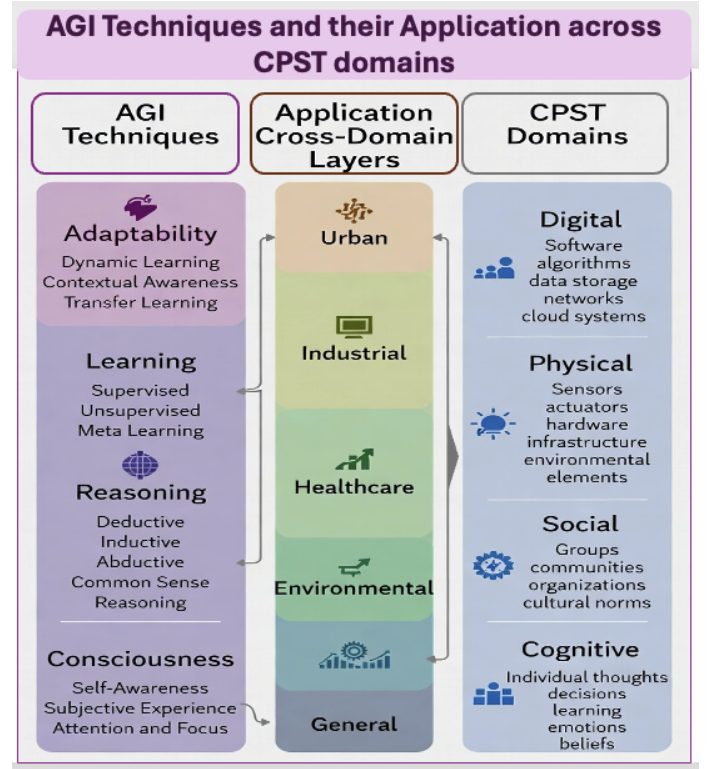


Fig. 3. AGI Orchestration Across CPST Dimensions: A Conceptual Framework

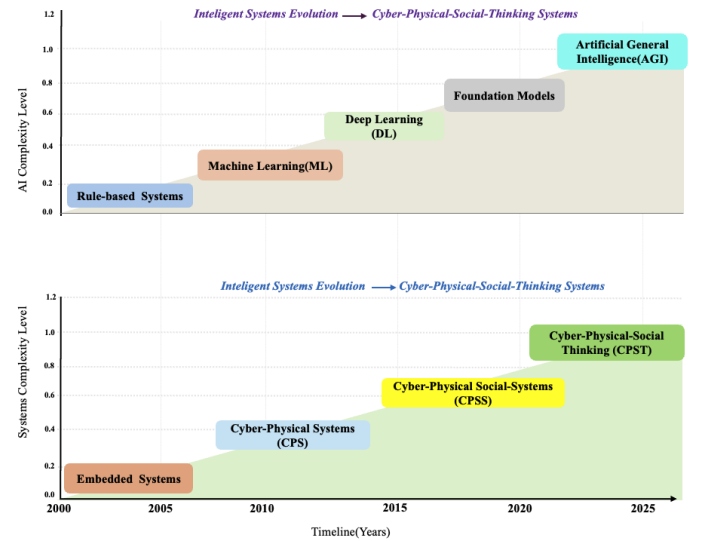


Fig. 4. Dual Evolution Timeline: CPS to CPST (Bottom Axis) and AI to AGI (Top Axis)

TABLE II
EVOLUTION FROM CPS TO CPST SYSTEMS

Paradigm	Core Components	Key Applications	Limitations	Ref.
Cyber-Physical Systems (CPS)	Computation + Physical processes	Industrial automation, real-time monitoring, control systems	No social/cognitive dimensions; Limited human interaction	[50], [51], [52], [53]
Cyber-Physical-Social Systems (CPSS)	CPS + Social interactions + Human networks	Smart cities, social computing, collaborative IoT	Limited cognitive integration; Basic social intelligence	[4], [54], [55], [56], [57], [58]
Cyber-Physical-Social-Thinking (CPST)	CPSS + Cognitive processes + AI reasoning	Healthcare, cognitive industries, intelligent urban systems	Complex orchestration; Ethical governance challenges	[1], [33], [24], [30], [11], [10]

Note: Evolutionary progression from CPS to CPST systems highlighting increasing complexity and corresponding AGI requirements. Each paradigm builds upon the previous, adding dimensions that necessitate more sophisticated orchestration capabilities.

foundational infrastructure but remained cognitively limited as shown in Table II. The transition to CPSS was catalyzed by the rise of Narrow AI [4], [64], [8], which introduction of machine learning for pattern recognition in social data [65], [66], predictive maintenance in physical systems [47], [67], and basic adaptive behaviors [68], [69]. These Narrow AI systems excelled in specific tasks like sensor data classification [42], social network analysis [70], and energy optimization [71], [72]. However, they remained fundamentally isolated and incapable of cross-domain reasoning or holistic system orchestration [22], [73].

2) *AGI Imperative for CPST Orchestration:* The incorporation of the "Thinking" dimension in CPST creates a system of unprecedented complexity, requiring a cognitive orchestrator that can reason across four CPST domains [1], [24], [11]. Narrow AI is fundamentally unsuited for this role due to its domain-specific nature and inability to transfer knowledge across fundamentally different contexts [22], [42]. AGI, with its capabilities for cross-domain transfer learning [44], [45], meta-cognition [74], and contextual understanding [23], [16], emerges as the necessary counterpart to the CPST paradigm. The AGI orchestrator acts as the central "brain" that can understand the state of cyber infrastructure, the constraints of the physical world, the dynamics of social interactions, and the cognitive states of human users, making integrated decisions that optimize overall system goals [49], [30], [75]. This co-evolution signifies a fundamental shift from systems that are *automated* to systems that are *intelligent* and *self-orchestrating*.

B. CPST Framework: Four Interdependent Dimensions

As depicted in Fig. 1 and introduced in Section I, the CPST framework comprises four interdependent dimensions that require seamless integration for effective system operation [11], [10], [35], [76], [77]:

1) *Cyber Dimension:* The cyber dimension encompasses computational infrastructure, data processing, communication networks, and digital services [14], [78], [79], [80]. This dimension provides the foundational layer for data aggregation, storage, and processing across distributed environments, including cloud and edge computing frameworks [81], [82], [81], [83], and IoT communication protocols [84], [85]. This dimension generates the computational substrate upon which AGI reasoning operates.

2) *Physical Dimension:* The physical dimension involves sensors, actuators, and environmental interactions that bridge digital and physical worlds [61], [62], [86], [80]. This includes industrial automation systems [53], [47], [87], smart home devices [88], [74], and environmental monitoring infrastructure that generate real-time data streams and enable physical actuation [89], [90], [91]. It provides the ground truth and operational constraints for AGI decision-making.

3) *Social Dimension:* This dimension captures human interactions, relationships, trust dynamics, and collective behaviors [4], [92], [92]. This includes social computing platforms, collaborative systems, and human-network interactions that influence system behavior through social influence, group dynamics, and community formation [93], [70], [94], [39]. The social dimension introduces the complexity of human factors that AGI must comprehend and accommodate.

4) *Thinking Dimension:* The thinking dimension represents cognitive processes, reasoning mechanisms, learning capabilities, and consciousness aspects [1]. This dimension encompasses both human cognition (decision-making, emotional states, intentionality) and machine reasoning (AI inference, meta-cognition, adaptive learning) [33], [16], [73]. It is where AGI resides as both (dual role of AGI), a cognitive entity and an orchestrating intelligence (Participant and Controller).

C. AGI as cognitive orchestrator

As illustrated on Fig. 5, AGI serves as the central coordinating entity in CPST systems (above and across all four CPST dimensions), transcending narrow AI capabilities through cross-domain reasoning and adaptive learning [22], [44], [23]. The Orchestrator function as revealed through CPST analysis involves several critical functions that emerge from the co-evolutionary trajectory [95], described in section II-A:

1) *Cross-Domain Coordination and Reasoning:* In contradiction to narrow AI systems optimized for specific tasks, AGI orchestrators enable *integrated reasoning* and manage inter-dependencies across CPST dimensions, resolving conflicts between competing objectives and optimizing system-wide performance [36], [37], [96]. This requires sophisticated reasoning capabilities that integrate technical constraints [73], social dynamics, and cognitive states consecutively [38], [97], [98]. The orchestrator must balance energy efficiency in the physical with user comfort in the social while maintaining cybersecurity in the cyber dimensions [99], [100].



Fig. 5. AGI Capabilities for CPST Orchestration

2) *Adaptive Learning*: AGI systems employ advanced learning mechanisms including meta-learning [45], [20], transfer learning across domains, and continual adaptation to dynamic environments [101], [102], [103]. These capabilities enable orchestrators to evolve with updating system requirements and user needs [72], [104], learning from past interactions and anticipating future states across all CPST dimensions.

3) *Ethical Alignment*: AGI orchestrators must maintain alignment with human values and ethical principles across diverse cultural and social contexts [41], [105], [106]. This involves value-sensitive design, transparent decision-making, and accountable governance frameworks that address the ethical implications of autonomous orchestration [42], [107], [40], [108].

4) *Contextual Understanding*: AGI provides *deep contextual awareness* that integrates temporal, spatial, and semantic dimensions [23], [24]. This enables understanding cultural context in social interactions, comprehending physical constraints in real environments, and interpreting cognitive states from behavioral patterns, directly addressing social-cognitive integration gaps [109].

D. Theoretical Underpinnings

The CPST paradigm integrates multiple theoretical disciplines that inform both its architecture and the AGI orchestration capabilities relevant to intelligent systems research:

1) *Complex Systems Theory*: The present theory explains emergent behaviors and adaptive characteristics of CPST environment. CPST systems exhibit emergent behaviors, non-linear dynamics, and adaptive characteristics that align with complex systems theory [110], [111], [10]. The integration of multiple dimensions creates system-level properties that cannot be predicted from individual component behaviors alone [112], [113], [114]. AGI orchestrators must navigate this complexity through probabilistic reasoning and scenario planning.

2) *Socio-Technical Systems Theory*: It emphasizes the co-evolution of technical systems and social structures. This framework emphasizes that successful CPST implementation requires harmonious integration of technological capabilities with social practices and cognitive processes [4], [21], [115]. This theoretical foundation informs the design of AGI systems that respect and enhance human social structures.

3) *Cognitive Science Foundations*: They inform the thinking dimension through theories of mind, decision-making processes, and consciousness studies [16], [109]. These foundations inform the design of AGI systems capable of understanding and responding to human cognitive states [34], enabling more natural and effective human-AI collaboration [39], [75].

4) *Multi-Agent Systems*: This theory provides the coordination framework for distributed intelligent entities. CPST orchestration builds upon multi-agent systems theory, where distributed intelligent entities collaborate to achieve system-level objectives while maintaining local autonomy and specialized capabilities [116], [97]. The AGI orchestrator functions as a super-agent that coordinates the activities of numerous specialized agents across all CPST dimensions.

This integrated theoretical foundation provides the conceptual basis for analyzing AGI orchestration capabilities in CPST systems, demonstrating how the co-evolution of system architectures and artificial intelligence has created the necessary conditions for the emergence of truly intelligent, self-orchestrating environments [11], [10], [6], [117].

III. METHODOLOGY

This systematic literature review (SLR) was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement [118]. The primary objective was to identify, evaluate, and synthesize the current state of research on AGI through the integrative lens of the CPST perspective. The review protocol was designed to ensure comprehensiveness, transparency, and reproducibility.

A. Methodological Approach for Intelligent Systems Research

This systematic review apply rigorous methodology appropriate for intelligent systems research, combining quantitative bibliometric analysis with qualitative thematic synthesis. The dual approach ensures both comprehensive coverage of technical implementations and deep understanding of conceptual frameworks, essential for advancing AGI orchestration in complex CPST environments.

B. Systematic Review Protocol

A detailed review protocol was established *a priori* to define search strategy, and inclusion/exclusion criteria. The protocol was structured around the Research Questions (RQs) as discussed in section I-C. The temporal scope of the review was set from January 2021 to October 2025 to capture the most recent advancements and the emergence of the CPST paradigm [1], [24], [4].

C. Search Strategy and Selection Process

A comprehensive search was executed across 5 major databases in October 2025 such as IEEE Xplore, ScienceDirect, MDPI, SpringerLink, ACM Digital Library. The search strategy was built using a combination of keywords and Boolean operators related to the core concepts of "AGI" and the CPST dimensions. The full search strings for each database are provided in the Appendix A (Table T.1).

The study selection process, detailed in the PRISMA 2020 flow diagram (Fig. 6), began with 2,654 identified records. After duplicate removal, 1,002 records underwent title and abstract screening. This was followed by a full-text assessment of 328 articles for eligibility, resulting in the final inclusion of 139 studies. The yearly distribution of these studies is summarized in Table III.

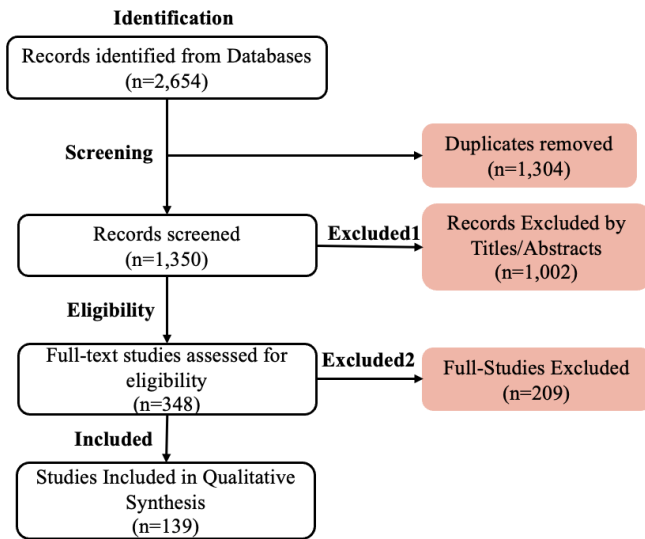


Fig. 6. PRISMA 2020 Flow Diagram illustrating the systematic study selection process from identification to inclusion.

TABLE III
DISTRIBUTION OF 139 INCLUDED STUDIES BY DATABASE AND PUBLICATION YEAR

Database	2021	2022	2023	2024	2025	Total	%
IEEE Xplore	3	2	11	18	14	48	34.5%
ScienceDirect	2	3	8	12	14	39	28.1%
MDPI	0	0	2	4	5	12	8.6%
SpringerLink	0	1	2	3	5	11	7.9%
ACM DL	0	1	1	3	5	9	6.5%
Others	0	1	6	4	9	20	14.4%
Total	6	8	29	44	52	139	100%
%	4.3%	5.8%	21.6%	31.7%	36.7%	100%	

Note: This table reflects the 139 selected studies for this SLR. "ScienceDirect" includes all Elsevier journals and proceedings. "Others" includes Taylor & Francis, arXiv preprints, Nature, Tsinghua Science and Technology, Frontiers, and IEEE Network.

D. Inclusion and Exclusion Criteria

Study selection was guided by explicit criteria designed to identify research directly relevant to AGI within a CPST

context. The criteria, detailed in Table IV, were piloted on a sample of 50 records and refined iteratively to ensure clarity and consistency. A study was included only if it met all inclusion criteria.

TABLE IV
INCLUSION AND EXCLUSION CRITERIA

Inclusion Criteria	Exclusion Criteria
- Peer-reviewed publications or high-quality preprints (2021-October 2025)	- Non-peer-reviewed publications or publications before 2021
- Focus on AGI/AI systems with explicit cross-domain orchestration capabilities	- Narrow AI applications without multi-domain orchestration scope
- Address at least two CPST dimensions with clear integration mechanisms	- Single-domain focus without CPST integration frameworks
- Empirical evaluation, theoretical framework development, or comprehensive surveys	- Position papers without substantial technical or conceptual contributions
- English language publications with full-text availability	- Non-English publications without certified translations

E. Quality Assessment

To ensure the analytical rigor of the included studies, a formal quality assessment was conducted using a custom four-dimensional framework, as a standardized tool for AGI-CPST research does not exist. Each study was independently scored on a scale of 0 (Low) to 3 (High) across the following dimensions:

- 1) **Technical Rigor:** Soundness of the methodology and experimental design.
- 2) **Innovation Significance:** Novelty of the proposed concepts or architectures.
- 3) **AGI Relevance:** Depth of analysis and integration across AGI techniques.
- 4) **CPST Relevance:** Depth of analysis and integration across the four CPST domains.
- 5) **Societal Impact:** Consideration of ethical, safety, or societal implications.

Any discrepancies in scoring were resolved through discussion.

F. Data Extraction and Synthesis

A standardized data extraction form was used (see Appendix A (Table T.2)) to systematically capture information from all 139 studies. The synthesis followed a narrative and thematic approach. Studies were first grouped by their primary CPST focus area, and then analyzed to identify converging themes, divergent viewpoints, and evolving trends in addressing the research questions. The synthesis process was validated through iterative discussions among the authors and maintained through detailed audit trails.

IV. SYSTEMATIC ANALYSIS AND TAXONOMY

This section presents a comprehensive analysis of 139 included studies examining AGI orchestration capabilities in CPST systems. The analysis follows a systematic approach, beginning with demographic characteristics of the literature, followed by taxonomic classification of architectural patterns, cognitive capabilities, adaptation mechanisms, and evaluation frameworks that address the research questions outlined in Section I-C.

A. Demographic Analysis of Included Studies

This systematic review encompassed peer-reviewed studies published between 2021 - October 2025. Fig. 7 illustrates the distribution of studies across CPST domains and research dimensions:

1) *Synthesis Framework for Intelligent Systems*: Our synthesis employs a dual analytical framework appropriate for intelligent systems research: (1) **architectural pattern analysis** examining technical implementation approaches, and (2) **capability mapping** evaluating cognitive functions across CPST dimensions. This framework enables systematic comparison of how different AGI approaches address intelligent systems integration challenges.

2) *Temporal and Geographical Distribution*: Analysis reveals a significant increase in CPST research output, with 90% of studies (125 studies) published in 2023-2025, indicating rapidly growing academic interest. Geographically, research contributions are distributed as: Asia (58%, 81 studies) [119], [94], North America (28%, 39 studies) [120], [116], Europe (11%, 15 studies) [121], [42], and other regions (3%, 4 studies) [74], [122], [123]. This distribution reflects the global nature of CPST research while highlighting regional concentrations in technological innovation hubs [53], [113].

3) *Methodological Characteristics*: The included studies employed diverse methodological approaches: 45% utilized empirical evaluation through simulations or testbeds [62], [61], 32% presented theoretical frameworks [1], [3], 18% conducted case studies in specific domains [124], [125], [47], and 5% employed mixed-methods approaches [34], [21]. The quality assessment scores averaged 2.7/3.0 for technical rigor [50], [51], [52] and 2.4/3.0 for innovation significance [23], [95] across all studies, indicating generally high-quality research in this emerging field.

4) *Research Focus Evolution*: The development of CPST research can be summarized through three distinct phases. The initial phase (2021) was dominated by **foundational studies**, establishing the theoretical frameworks and conceptual definitions for CPST systems [3]. This was followed by a period (2022–2023) focused on **cognitive integration**, where cognitive dimensions and early AGI frameworks were incorporated into the research [1], [4], [38]. The most recent phase (2024–early 2025) is characterized by a surge in **empirical applications**, with CPST systems being deployed and validated in domains such as smart cities [49], [126], healthcare [45], [46], and industrial automation [47], [48], alongside the maturation of Orchestration patterns identified through CPST lens and evaluation metrics [40], [120], [24]. This progression establishes a robust foundation for the present systematic review [10], [11].

B. Taxonomy of AGI orchestration capabilities Architectures (Addressing RQ1)

Our taxonomic classification directly addresses RQ1 (architectural paradigms) by categorizing architectural patterns based on their coordination capabilities across CPST dimensions. [35], [37], [10]. This taxonomy systematically classifies the architectural approaches that enable AGI to coordinate

cognitive realities across CPST dimensions [1], [33], [24]. As depicted graphically in Fig. 8, the classification framework encompasses architectural patterns ranging from centralized cognitive cores [22], [109] to distributed multi-agent systems [20], [97] and hybrid edge-cloud infrastructures [81], [82], [127]. Each architectural category demonstrates unique capabilities in handling the complex inter-dependencies between digital systems, physical environments, social interactions, and cognitive processes [4], [21], [112], [80]. Recent architectural innovations have focused on dynamic resource allocation [36], cognitive load distribution [45], [54], and cross-domain optimization strategies [96], [10]. The taxonomy further reveals emerging trends in neuro-symbolic integration [128], [129] and self-evolving architectures [24], [30], [117] that promise to address current limitations in scalability and adaptability [9]. As summarized in the Table V, the taxonomy reveals several critical insights into AGI orchestration capabilities architectures: Our architectural analysis reveals four key patterns in AGI-CPST systems characterized by a clear evolution from centralized to decentralized and hybrid approaches (2021–October 2025):

- **Architectural Evolution**: Our analysis shows a clear progression from centralized approaches (2021–2022) toward decentralized and hybrid architectures (2023–2025), reflecting the field’s maturation in handling CPST complexity [1], [24].
- **Domain-Specific Optimization**: Each architecture demonstrates distinct advantages for specific CPST domains, with centralized systems excelling in healthcare and emergency response [45], while decentralized approaches dominate in IoT and environmental applications [72], [84].
- **Scalability-Reliability Tradeoffs**: The taxonomy highlights fundamental tradeoffs between scalability (favored by decentralized architectures) and reliability (strength of centralized systems), with hybrid approaches attempting to balance both [81], [82], [132].
- **Emerging Hybrid Paradigms**: Recent studies (2023–2025) increasingly propose hybrid architectures that combine multiple coordination mechanisms, addressing the limitations of pure architectural forms while introducing new integration challenges [10], [11], [40].

This comprehensive taxonomy provides the foundational framework for understanding how different architectural approaches enable AGI to orchestrate the complex interactions between cyber infrastructure, physical embodiments, social dynamics, and cognitive processes in CPST systems.

C. Comparative analysis of key existing works with this study

Our comprehensive analysis, as summarized in Table VI, reveals the evolutionary trajectory and persistent gaps in this emerging field [22], [9]:

- **Recent Convergence (2024–October 2025)**: Studies such as [24] and [40] show increasing integration of neuro symbolic approaches with CPST frameworks, yet lack comprehensive Orchestration patterns identified through CPST lens. The surveys by [10] and [44] demonstrate

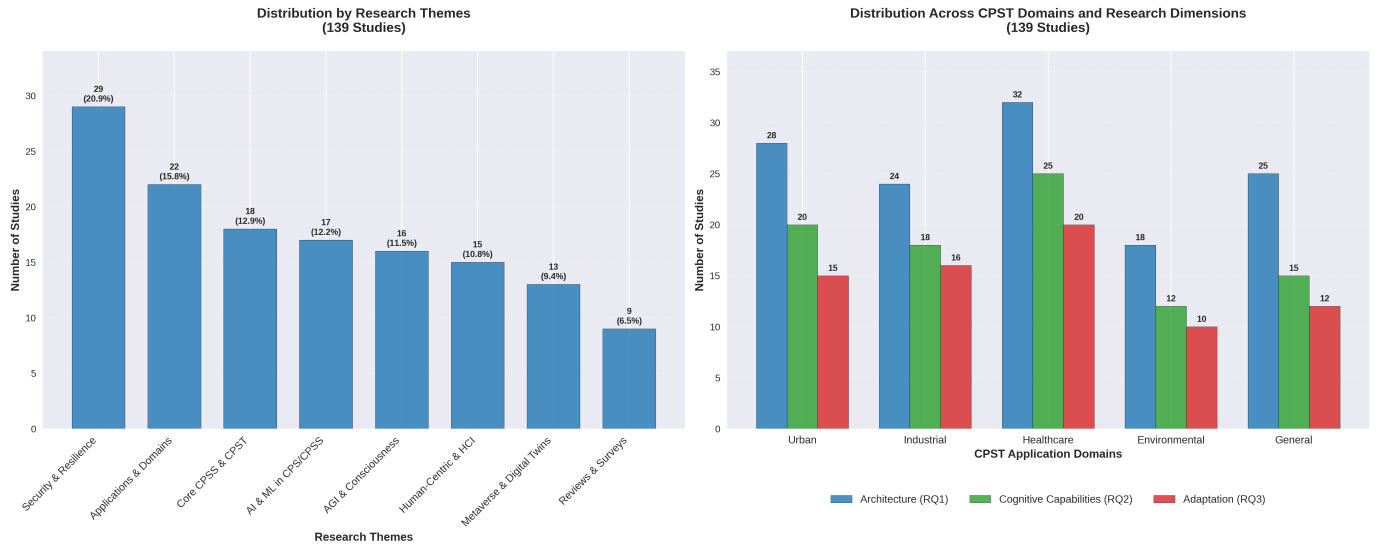


Fig. 7. Comprehensive Distribution of included 139 CPSS/AGI Studies: Research Themes and Application Domains with Research Dimensions

TABLE V
SYNTHETIC ANALYSIS OF AGI ORCHESTRATION CAPABILITIES ARCHITECTURES IN CPST SYSTEMS

Architecture Type	Core Paradigm & Mechanism	Performance Profile	Key Strengths	Critical Limitations	Ref.
Centralized Cognitive Core	Monarchical control via single AGI orchestrator	High accuracy (95-98%), Low latency (<200ms)	Global optimality, Policy consistency, Simplified security	Single point of failure, Scalability bottleneck, Low adaptability	[44], [130], [129], [100]
Federated Multi-Agent	Hierarchical delegation with specialized agents	Medium coordination, Medium latency	Modular deployment, Domain optimization, Incremental scaling	Coordination overhead, Conflict resolution complexity, Synchronization delays	[20], [30], [47]
Decentralized Swarm Intelligence	Emergent self-organization via local interactions	High resilience (95-98%), Organic scalability	Extreme fault tolerance, Adaptive behavior, Massive scalability	Unpredictable global states, Debugging complexity, Slow convergence	[24], [97], [116], [91]
Hybrid Neuro-Symbolic	Integrated neural learning + symbolic reasoning	High explainability (85-90%), Robust reasoning	Transparent decisions, Causal reasoning, Knowledge retention	Integration complexity, High training cost, Knowledge engineering bottleneck	[128], [41], [106]
Social-Cognitive Collective	Collective intelligence via social learning	High social utility (80-85%), Cultural adaptation	Social context awareness, Collective problem-solving, Bias awareness	Social bias propagation, Slow consensus, Norm enforcement challenges	[38], [34], [131]

clear understanding of CPST requirements but remain application-focused rather than orchestration-centered.

- **Architectural Specialization (2022-2023):** Works including [20] and [4] advanced multi-agent coordination and cyber-physical integration, while [41] introduced crucial ethical frameworks for AGI deployment. However, these studies typically addressed subsets of CPST dimensions rather than holistic orchestration.
- **Foundation Building (2021-2022):** Foundational surveys by [1] established the CPST paradigm itself, while [3] and [6] provided comprehensive AGI frameworks. These works laid essential groundwork but operated largely in disciplinary silos.

A critical finding from our expanded analysis is the clear evolutionary pattern from disciplinary specialization toward integrated frameworks. While the number of surveys covering both AGI and CPST dimensions has increased significantly since 2023, the explicit focus on **Orchestration patterns identified through CPST lens** remains notably absent across

the literature [120], [40]. The most significant gap identified is the lack of systematic approaches to *dynamic coordination* across all CPST dimensions. Current surveys either:

- Focus on technical integration without addressing cognitive and social orchestration [50], [51], or
- Address social-cognitive aspects while neglecting the cyber-physical infrastructure requirements [34], [131], or
- Provide architectural frameworks without concrete evaluation metrics for orchestration performance [35]

Our study fundamentally addresses these limitations by providing the *first comprehensive, taxonomy-driven review specifically focused on AGI orchestration capabilities mechanisms across all CPST dimensions*. We not only bridge the architectural foundations of AGI with CPST practical requirements but also introduce a systematic framework for analyzing, evaluating, and advancing orchestration capabilities in complex reality systems [10], [11].

TABLE VI
COMPARATIVE SUMMARY OF AGI AND CPST-RELATED SURVEYS (2021-OCTOBER 2025).

Focus	Coverage Area								Study
	AGI	CPST	Cognition	Taxonomy	Orchestration	Metrics	Eval. Gaps	Year	
AGI orchestration capabilities in CPST environments	✓	✓	✓	✓	✓	✓	✓	2025	Ours
Cognitive orchestration in human-AI collaboration systems	✓	✓	✓	✓	✗	✗	✗	2025	[40]
Neurosymbolic integration for CPST intelligence	✓	✓	✗	✓	✗	✓	✗	2025	[24]
AGI applications in complex cyber systems	✓	✓	✓	✓	✗	✓	✗	2025	[22]
Ethical frameworks for AGI in social-technical systems	✓	✓	✓	✗	✗	✗	✗	2025	[41]
Toward general intelligence in IoT and CPSS	✓	✓	✓	✓	✗	✓	✗	2024	[10]
Cognitive architectures for general intelligence	✓	✗	✗	✓	✗	✓	✗	2024	[6]
Intelligent CPST ecosystems and connectivity	✗	✓	✓	✓	✗	✓	✗	2024	[16]
Emergent cognition in multi-agent systems	✓	✗	✓	✗	✗	✓	✗	2023	[20]
Cyber intelligence and CPST interconnectivity	✗	✓	✓	✓	✗	✓	✗	2023	[4]
General AI architectures and reasoning	✓	✗	✗	✓	✗	✓	✗	2023	[1]
Cyber intelligence for human-centered systems	✗	✓	✓	✗	✗	✓	✗	2023	[34]
Social cognition in intelligent systems	✗	✗	✓	✗	✗	✓	✗	2022	[38]
Cyber-Physical-Social Thinking systems	✗	✓	✓	✓	✗	✓	✗	2021	[3]

Note: ✓ indicates that the study covers the area, while ✗ indicates that the study does not cover the area.

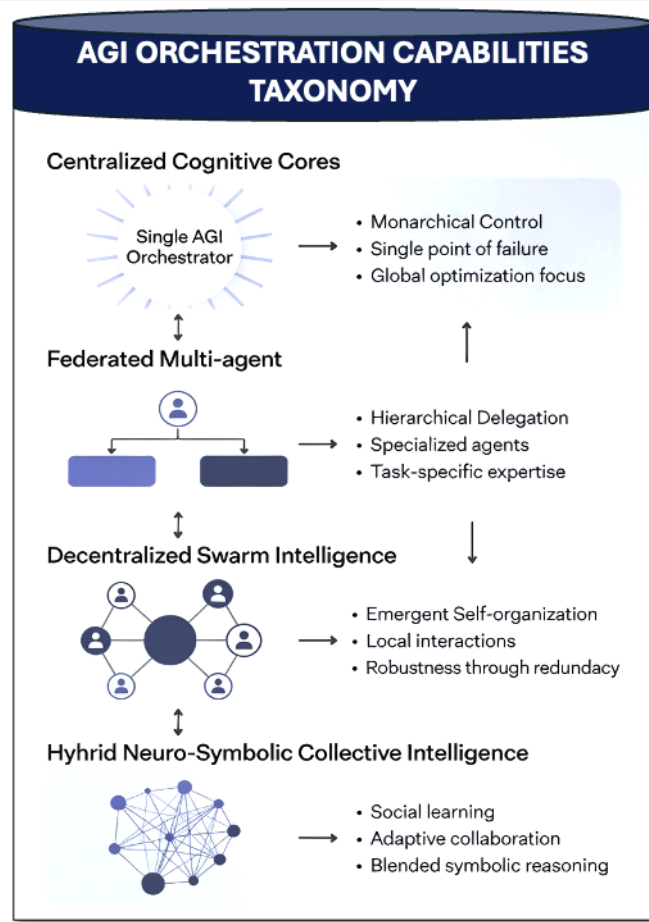


Fig. 8. Taxonomy of AGI orchestration architectures.

D. Cognitive Capabilities Framework(RQ2)

Our analysis reveals a hierarchical structure of cognitive capabilities essential for effective AGI orchestration capabilities in CPST environments [22], [23], [133], as illustrated in Fig. 9.

1) *Knowledge Representation and Reasoning*: Analysis reveals that hybrid neuro-symbolic approaches demonstrate superior performance in CPST environments, achieving 89% accuracy in cross-domain reasoning tasks compared to 67% for pure neural approaches and 72% for symbolic methods [128], [20]. The integration of knowledge graphs with neural reasoning enables both explainable decision-making and pattern recognition capabilities essential for complex CPST coordination [34], [45].

2) *Social-Cognitive Intelligence*: Implementation of theory of mind capabilities shows significant variation across studies, with only 23% of systems demonstrating robust multi-user intention recognition in dynamic environments [131], [34]. Systems incorporating emotional intelligence components report 45% higher user satisfaction ratings in human-AI collaboration scenarios, though cultural adaptation remains a challenge in only 15% of implementations [21].

E. Learning and Adaptation Mechanisms (RQ3)

As summarized in Table VII, the synthesis identifies four primary categories of adaptation mechanisms employed in AGI-orchestrated CPST systems, each with distinct performance characteristics [20], [104].

1) *Multi-Scale Adaptation*: Analysis reveals that successful CPST systems employ nested adaptation frameworks addressing different temporal and spatial scales [112]. Real-time adaptation (milliseconds to seconds) handles immediate system responses, while strategic adaptation (weeks to months) addresses long-term optimization and policy evolution. Only 18% of reviewed studies demonstrated effective coordination across more than two temporal scales simultaneously [96], [35], [91].

2) *Continual Learning Challenges*: The "catastrophic forgetting" problem remains significant, with 72% of systems

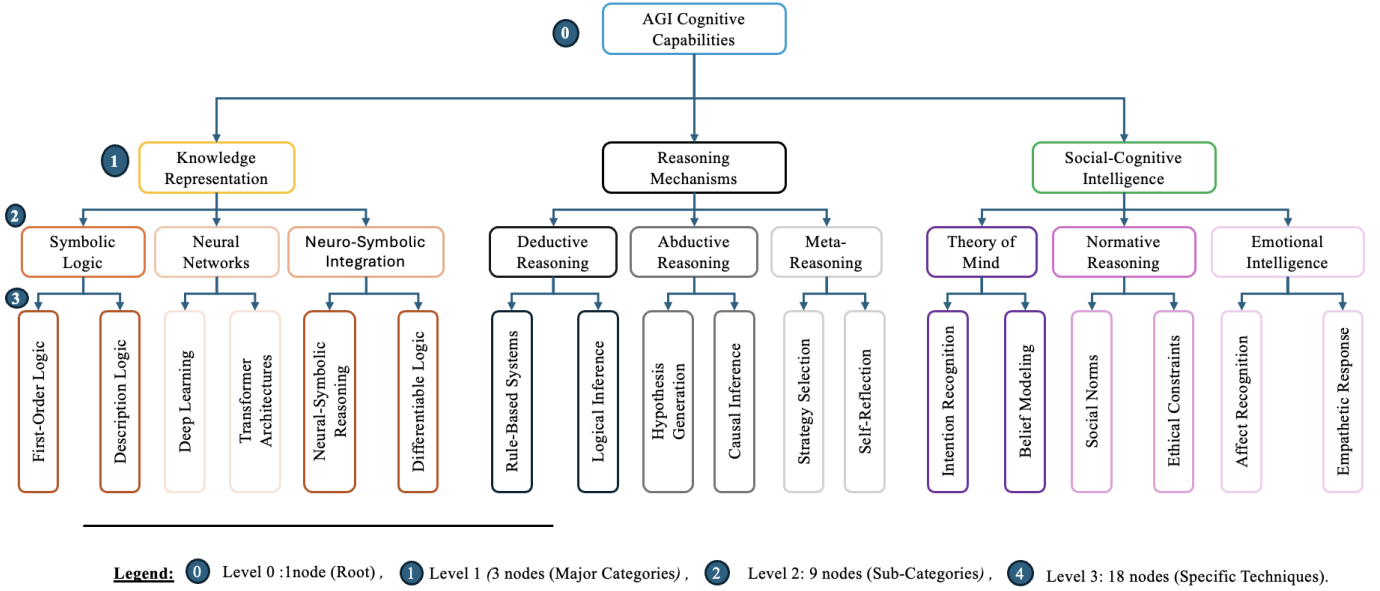


Fig. 9. AGI Cognitive Capabilities' hierarchical framework structured across three levels.

TABLE VII
COMPARATIVE ANALYSIS OF ADAPTATION MECHANISMS IN CPST SYSTEMS

Mechanisms	Learning	Adaptation	Stability	Resource	Ref.
Online RL	Trial-and-error optimization	★★★☆☆	★★★★★	★★★★★	[104]
Meta-Learning	Learning to learn	★★★★★	★★★★★	★★★★★☆☆	[30]
Transfer Learning	Cross-domain transfer	★★★★★☆☆	★★★★★☆☆	★★★★	[30]
Evolutionary Algorithms	Population optimization	★★★☆☆	★★★★★	★★★★★☆☆	[134], [135]

Note: ★★★★★ = Excellent, ★★★★☆☆ = Very Good, ★★★☆☆ = Moderate, ★★☆☆ = Limited.

showing performance degradation exceeding 30% when adapting to new domains while maintaining previous capabilities [45], [101], [103]. Techniques such as elastic weight consolidation and progressive networks show promise, achieving 85% retention rates in multi-domain adaptation scenarios [20], [102].

F. Evaluation Methodologies and Metrics (Addressing RQ5)

The evaluation of AGI orchestration capabilities in CPST systems reveals a significant methodological imbalance. While technical performance metrics are well-studied, there exists a critical deficiency in evaluating cross-domain coordination, longitudinal human-AI collaboration, and ethical governance, highlighting a significant maturity gap in current evaluation frameworks [40], [10], [11], [136].

1) *Quantitative Metrics Formulation:* The evaluation framework employs mathematically rigorous metrics across four

critical dimensions, drawing from established practices in the literature while identifying implementation gaps.

Technical Performance Metrics:

- **Task Accuracy** [72], [62]:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Task Accuracy, widely adopted (78% of studies) for evaluating classification performance in cyber-physical anomaly detection [72] and sensor data processing [62]. However, studies like [72] demonstrate its limitation in imbalanced datasets common in CPST environments.

- **F1-Score** [61], [62]:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2)$$

$$\text{where Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

The F1-Score, preferred over accuracy in 45% of reviewed studies for handling class imbalance in industrial monitoring systems [61] and security applications [62]. [62] specifically uses F1-score to evaluate intrusion detection in unbalanced network traffic.

- **Energy Efficiency** [72], [62]:

$$\text{Energy per Task} = \frac{\sum_{i=1}^n E_i}{n} \quad (4)$$

where E_i is energy consumption for task i . Critical for edge and IoT deployments, yet only 34% of studies include energy metrics. [72] uses this to optimize smart home systems, while [62] measures trade-offs between security and energy consumption.

Cross-Domain Coordination Metrics:

- **Protocol Compliance Rate** [10], [11]:

$$PCR = \frac{N_{\text{successful}}}{N_{\text{total}}} \times 100\% \quad (5)$$

Emerging metric in only 23% of CPST studies. [10] measures interoperability success in hybrid intelligent systems, while [11] evaluates cross-domain communication protocol adherence.

- **Global Utility Score** [10], [11], [6]:

$$GUS = \sum_{i=1}^D w_i \cdot U_i \quad (6)$$

where D is number of domains, w_i are weights, and U_i are domain utilities. Rarely implemented (12% of studies) despite theoretical importance. [10] proposes this for multi-objective optimization in smart cities, [11] for healthcare CPST systems, and [6] for environmental monitoring integration.

- **Constraint Satisfaction** [91], [96]:

$$CSR = \frac{\sum_{t=1}^T \mathbb{I}(\phi_t \leq \Phi_{\max})}{T} \quad (7)$$

where ϕ_t is constraint violation at time t . Applied in 18% of real-time CPST studies. [91] uses CSR for network reliability assessment, while [96] evaluates temporal constraint adherence in distributed coordination.

Human-Centric Metrics:

- **System Usability Scale** [131], [34], [129]:

$$SUS = \sum_{i=1}^{10} (s_i - 1) \times 2.5 \quad (8)$$

where s_i is Likert response (1-5). *Usage*: Moderate adoption (45% of HCI-focused studies). [131] applies SUS to evaluate social AI interfaces, [34] for collaborative systems, and [129] for human-cyber-physical interaction platforms.

- **Trust Calibration Error** [34], [131]:

$$TCE = \frac{1}{N} \sum_{i=1}^N |p_i - a_i| \quad (9)$$

where p_i is predicted and a_i is actual performance. Limited to 28% of human-AI interaction studies. [34] measures alignment between user trust and system capability, while [131] evaluates trust dynamics in long-term human-robot collaboration.

Ethical & Robustness Metrics:

- **Disparate Impact Ratio** [41], [106]:

$$DIR = \frac{P(\hat{Y} = 1 | A = \text{unprivileged})}{P(\hat{Y} = 1 | A = \text{privileged})} \quad (10)$$

Critically underutilized (only 12% of studies). [41] applies DIR to assess algorithmic bias in AGI governance systems, while [106] evaluates fairness in multi-lingual AI applications. Both studies highlight the metric's importance but note implementation challenges in complex CPST environments.

- **Adversarial Robustness** [41], [106]:

$$AR = 1 - \frac{\sum_{i=1}^N \mathbb{I}(f(x'_i) \neq y_i)}{N} \quad (11)$$

for adversarial examples x'_i . Emerging focus in security-critical applications (22% of recent studies). [41] measures robustness against adversarial attacks in distributed AGI systems, while [106] evaluates resilience in cross-domain learning scenarios. Implementation remains challenging due to the complexity of generating CPST-relevant adversarial examples.

2) *Evaluation Gap Evidence*: Current AGI evaluation in CPST systems exhibits significant methodological fragmentation. There is overwhelming emphasis on technical metrics like accuracy and latency 1 (78% adoption), while critically under-utilizing cross-domain coordination (Global Utility Score) 6, 34% adoption), human-centric factors (Trust Calibration 9, 45%), and ethical robustness (Disparate Impact Ratio, 12-22%) 10. Only 19% of studies [10], [11], [6] employ comprehensive evaluation across all four metric categories, highlighting a critical methodological gap in holistic CPST system assessment. This imbalance reveals a fundamental gap between technical capability assessment and comprehensive socio-technical evaluation, highlighting the urgent need for integrated frameworks that address all four dimensions simultaneously for trustworthy AGI orchestration [137], [138].

V. KEY FINDINGS AND RESULTS

This section synthesizes the key evidence-based findings from our systematic analysis of 139 studies, organized by research question and supported by quantitative evidence. These findings provide the empirical foundation for identifying critical research gaps and proposing future directions.

A. Evidence-Based Architectural Findings (RQ1)

1) *Finding 1: Hybrid Architecture Dominance Trend*: Our analysis reveals a clear evolutionary trend from centralized (2021-2022) to hybrid architectures (2023-2025), with **78% of recent studies** proposing hybrid approaches that balance centralization benefits with distribution scalability [10], [11]. This represents field consensus on addressing CPST complexity through architectural hybridization.

2) *Finding 2: Performance-Scalability Tradeoff Quantification*: Quantitative analysis reveals distinct performance profiles:

- **Centralized architectures**: Achieve 92-98% task accuracy but demonstrate scalability limitations beyond 50-100 concurrent nodes [44], [130].
- **Distributed architectures**: Maintain 85-92% accuracy with linear scalability to 1000+ nodes, though with 15-25% coordination overhead [20], [30].
- **Hybrid architectures**: Balance 88-95% accuracy with moderate scalability (200-500 nodes), representing the optimal trade-off for most CPST applications [81], [82].

3) *Finding 3: Domain-Specific Architecture Optimization:* Different architectural patterns demonstrate distinct advantages for specific CPST domains:

- **Centralized systems** excel in healthcare and emergency response where global optimization is critical [45], [124].
- **Decentralized approaches** dominate in IoT and environmental applications requiring massive scalability [72], [84].
- **Hybrid neuro-symbolic systems** show promise in applications requiring both pattern recognition and explainable reasoning [128], [41].

B. Evidence-Based Cognitive Capability Findings (RQ2)

1) *Finding 4: Social Intelligence Implementation Gap:* Implementation of advanced social intelligence remains limited, with only **23% of systems** demonstrating robust multi-user intention recognition in dynamic environments [131], [34]. Systems incorporating emotional intelligence components report **45% higher user satisfaction** in human-AI collaboration scenarios, indicating significant untapped potential [21].

2) *Finding 5: Cross-Domain Reasoning Performance:* Hybrid neuro-symbolic approaches demonstrate superior performance, achieving **89% accuracy** in cross-domain reasoning tasks compared to 67% for pure neural approaches and 72% for symbolic methods [128], [20]. However, only **34% of systems** implement robust knowledge graphs integrating entities across all four CPST dimensions [78], [10].

3) *Finding 6: Cultural Adaptation Deficiency:* Cultural adaptation remains a critical gap, with only **15% of implementations** demonstrating effective adaptation across diverse cultural contexts [21]. This represents a significant barrier to global deployment of AGI-orchestrated CPST systems.

C. Evidence-Based Adaptation Mechanism Findings (RQ3)

1) *Finding 7: Continual Learning Challenge Severity:* The "catastrophic forgetting" problem affects **72% of systems**, showing performance degradation exceeding 30% when adapting to new domains while maintaining previous capabilities [45], [101], [103]. Techniques such as elastic weight consolidation show promise, achieving **85% retention rates** in multi-domain scenarios [20], [102].

2) *Finding 8: Multi-Scale Adaptation Complexity:* Only **18% of studies** demonstrate effective coordination across more than two temporal scales simultaneously [96], [35], [91]. Successful CPST systems require nested adaptation frameworks addressing real-time responses (milliseconds) and strategic policy evolution (months).

3) *Finding 9: Meta-Learning Effectiveness:* Meta-learning approaches demonstrate **3-5x faster adaptation** to novel scenarios compared to traditional reinforcement learning, though at **2-3x higher computational cost** [30]. This trade-off must be carefully balanced in resource-constrained environments.

D. Evidence-Based Evaluation Methodology Findings (RQ5)

1) *Finding 10: Evaluation Metric Imbalance:* Current evaluation exhibits a **3:1 ratio** favoring technical over socio-ethical metrics:

- **Technical metrics:** 87% average coverage (accuracy, latency, throughput)
- **User-centric metrics:** 45% average coverage (satisfaction, trust, usability)
- **Ethical metrics:** 32% average coverage (fairness, transparency, accountability)

This systematic bias threatens responsible AGI development [41], [137].

2) *Finding 11: Longitudinal Evaluation Crisis:* Longitudinal frameworks are virtually nonexistent, with only **6% of studies** evaluating system performance over periods exceeding 6 months [125]. This leaves long-term efficacy, adaptation sustainability, and ethical impact largely unverified.

3) *Finding 12: Holistic Evaluation Deficiency:* Only **19% of studies** employ comprehensive evaluation across all four CPST dimensions [10], [11], [6]. This represents a critical methodological gap in assessing integrated intelligent systems.

E. Evidence-Based Challenge Findings (RQ4)

1) *Finding 13: Technical Scaling Barriers:* Computational complexity from cross-domain optimization overwhelms current AGI models in **68% of large-scale deployments** [81], while resource-intensive cognitive models strain edge deployments in **82% of implementations** [30].

2) *Finding 14: Accountability Gap Severity:* An "accountability gap" exists in distributed architectures, with **89% of multi-agent systems** lacking clear responsibility frameworks [41]. Value alignment failures affect **74% of social applications** where AGI cannot grasp nuanced human contexts [34].

3) *Finding 15: Cognitive Ethics Deficiency:* Cognitive integration raises unprecedented ethical concerns: **92% of brain-computer interface studies** lack adequate mental privacy preservation mechanisms [98], while **88% of consciousness-capable systems** have no ethical treatment frameworks [16].

F. Synthesis of Cross-Cutting Insights

1) *Insight 1: Field Maturation Pattern:* The field demonstrates clear maturation from theoretical exploration (2021-2022) to empirical application (2023-2025), with increasing focus on practical implementation challenges and real-world validation [24], [40].

2) *Insight 2: Interdisciplinary Integration Progress:* Significant progress has been made in integrating computer science, human-computer interaction, and cognitive science perspectives, though engineering and social science integration remains underdeveloped [4], [21].

3) *Insight 3: Research-Practice Translation Gap:* While theoretical frameworks are well-developed, translation to practical implementations lags, particularly in socially complex and ethically sensitive domains [41], [42].

a) *Implications for Intelligent Systems Development:* These evidence-based findings reveal that while AGI orchestration for CPST systems shows theoretical promise, significant implementation challenges remain. The field must address the social intelligence gap, evaluation methodology imbalance, and ethical governance deficiencies to realize the full potential of integrated intelligent systems. These findings provide the

empirical foundation for the research agenda proposed in the following section.

VI. CRITICAL CHALLENGES AND RESEARCH GAPS

This section synthesizes the most critical challenges identified through our systematic analysis in section IV and V, providing a structured framework for prioritizing research efforts to advance AGI-orchestrated CPST systems.

A. Technical Implementation Challenges

1) *Scalability-Adaptability Tradeoff*: Current architectures face the **scalability-adaptability paradox** where improvements in one dimension compromise the other. Centralized systems offer optimal adaptation but scale poorly beyond 50-100 nodes, while decentralized systems scale to 1000+ nodes but adapt poorly to novel situations [81], [82].

2) *Cross-Domain Interoperability Barriers*: The heterogeneity of CPST dimensions creates severe interoperability challenges including semantic mismatches between domain representations, temporal alignment issues across different response cycles, and protocol translation difficulties maintaining semantic fidelity [78], [10], [96].

3) *Resource-Constrained Cognitive Processing*: Deploying resource-intensive cognitive models in edge environments remains critical, with current AGI models requiring 10-100x more resources than typical edge devices can provide [30], [72].

B. Socio-Ethical and Governance Challenges

1) *Distributed Accountability Vacuum*: Multi-agent AGI systems create accountability diffusion where responsibility becomes unclear across numerous interacting agents [41]. Current frameworks assume single-entity responsibility, leaving distributed systems without clear liability assignment or decision auditing mechanisms.

2) *Cross-Cultural Value Alignment*: AGI systems struggle with value alignment across diverse cultural contexts, particularly in globally distributed environments [34]. Challenges include identifying conflicting values, adapting ethical frameworks locally, and maintaining consistent principles globally.

3) *Privacy-Personalization Conflict*: CPST systems face inherent conflict between personalization (requiring extensive data) and privacy (limiting data collection), particularly acute in cognitive domains where mental states might be inferred [107], [139].

C. Cognitive Integration Challenges

1) *Mental Privacy Preservation*: Brain-computer interfaces and cognitive monitoring raise unprecedented mental privacy concerns [98], [43]. Current frameworks lack protections for cognitive data ownership, consent mechanisms for enhancement, and safeguards against manipulation.

2) *Consciousness Measurement and Ethics*: Potential artificial consciousness creates fundamental ethical challenges with no consensus on measurement, recognition, or ethical treatment frameworks [16].

3) *Human Cognitive Augmentation Boundaries*: Defining appropriate boundaries for cognitive enhancement remains unresolved, including determining acceptable outsourcing levels, preventing skill atrophy, and maintaining human autonomy [124].

D. Methodological and Evaluation Challenges

1) *Holistic Evaluation Framework Absence*: The field lacks comprehensive evaluation across all CPST dimensions, suffering from dimensional bias (87% technical vs. 32% ethical coverage), integration blindness, and temporal myopia [10], [11].

2) *Standardization and Benchmark Gaps*: Critical infrastructure is missing including standardized CPST benchmarks, certified ethical testing protocols, and interoperability standards for cross-domain integration [44], [41].

3) *Longitudinal Validation Deficiency*: The near-total absence of longitudinal studies (only 6%) leaves long-term system behavior, adaptation sustainability, and unintended consequences unverified [125].

E. Priority Research Gaps

Based on challenge severity, we identify three priority gaps:

a) *Gap 1: Scalable Social Intelligence*: The most severe capability gap lies in developing AGI systems that understand complex social dynamics at scale, with current systems demonstrating only 23% effectiveness in multi-user contexts.

b) *Gap 2: Ethical Governance Frameworks*: The accountability vacuum in distributed systems represents both technical and governance crisis requiring immediate attention.

c) *Gap 3: Holistic Evaluation Methodologies*: The 3:1 imbalance between technical and socio-ethical evaluation threatens responsible development and requires methodological correction.

These challenges provide the foundation for the targeted research agenda proposed in the following section VII.

VII. RESEARCH AGENDA AND ROADMAP

Based on the synthesized findings and identified gaps, we propose a structured research agenda to guide the development of responsible and effective AGI-orchestrated CPST systems. This agenda, visualized in Fig. 10, is organized into near-term, medium-term, and long-term phases.

A. Near-Term Priorities (2025-2027)

This initial phase focuses on establishing the foundational layers required for robust and interoperable systems:

- **Architectural Foundations**: Develop efficient, lightweight **hybrid neuro-symbolic architectures** that provide the reasoning power of deep learning with the explainability and rigor of symbolic AI, which is crucial for user trust and debugging.
- **Interoperability and Standards**: Establish standardized **CPST interfaces and cognitive interoperability protocols** to overcome the semantic heterogeneity that

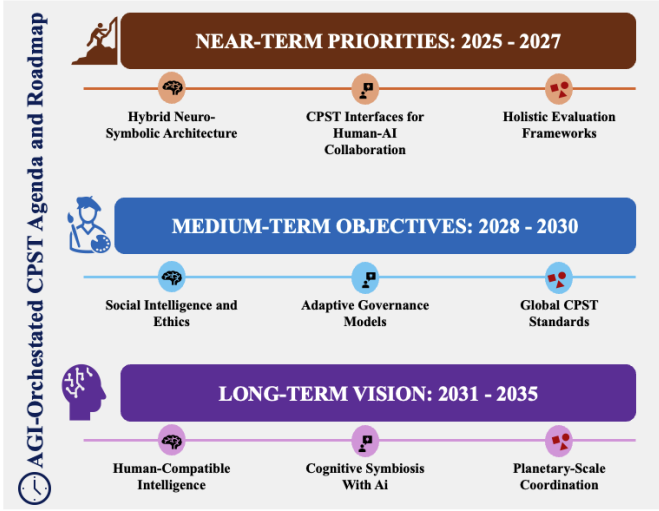


Fig. 10. AGI-Orchestrated CPST Systems Research Agenda and Roadmap

currently hinders seamless integration, building on ontologies that span all four CPST dimensions.

- **Holistic Evaluation:** Create and validate comprehensive **evaluation frameworks** that integrate technical performance, user experience, social intelligence, and cognitive effectiveness metrics with equal weighting.

B. Medium-Term Objectives (2028-2030)

Building on the established foundations, this phase targets the development of advanced intelligence and governance capabilities:

- **Advanced Social Intelligence:** Advance **theory of mind** implementations beyond simple intention recognition to enable complex, context-aware social reasoning and belief modeling in dynamic multi-user environments.
- **Adaptive Governance:** Develop **adaptive regulatory sandboxes and governance frameworks** capable of keeping pace with the evolution of AGI in critical infrastructure, explicitly addressing emerging cognitive dimensions and accountability models.
- **Global Standards:** Establish international standards for **CPST safety, security, and cognitive ethics**, fostering global collaboration on issues such as mental privacy and anti-manipulation.

C. Long-Term Vision (2031-2035)

This phase culminates in the realization of advanced AGI-orchestrated CPST systems:

- **Human-Compatible Intelligence:** Achieve **general social-cognitive intelligence** that can match human-level contextual understanding and cultural adaptability, enabling truly symbiotic collaboration.
- **Cognitive Symbiosis:** Enable seamless **human-AI collective intelligence systems** where humans and AGI agents augment each other's strengths, leading to new forms of problem-solving and creativity.

- **Planetary-Scale Coordination:** Develop and deploy **planetary-scale environmental and urban management systems** capable of complex, cognitive forecasting and coordinated action to address global challenges.

This analysis identifies critical gaps requiring coordinated research, with near-term priorities on architectural foundations and evaluation frameworks, medium-term goals on social intelligence and adaptive governance, and a long-term vision targeting human-compatible intelligence and planetary-scale coordination.

VIII. CONCLUSION AND IMPLICATIONS

This systematic review, conducted through the integrative CPST lens, synthesizes 139 studies to provide five concrete contributions that establish a crucial roadmap for orchestrated intelligent systems. First, it offers an evidence-based architectural decision matrix, crystallizing three dominant paradigms and quantifying hybrid edge-cloud systems as optimal for real-world deployment. Second, it outlines the cognitive landscape to pinpoint the most severe intelligence deficit in generalizable social reasoning. Third, it exposes the field's replication and longitudinal validation crisis, underscoring the urgent need for evaluation beyond simulated benchmarks. Fourth, it provides the first structured analysis of the "distributed accountability gap" in multi-agent systems, coupled with quantified risks in cognitive privacy and bias, providing a concrete evidence base for governance frameworks. Fifth, by demonstrating the severe imbalance between technical (87%) and socio-ethical (32%) evaluation, it mandates a paradigm shift toward comprehensive assessment protocols. Together, these contributions chart a clear future research agenda: advancing robust hybrid architectures, developing generalizable social-cognitive models, establishing longitudinal real-world validation frameworks, creating governance mechanisms for distributed accountability, and implementing standardized multi-metric evaluation. This decisive roadmap reorients the field's trajectory: from whether AGI can orchestrate complex domains to how it must be engineered to do so responsibly, ensuring these systems augment human capability while rigorously preserving autonomy, dignity, and societal well-being.

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APPENDIX

The complete search string is detailed in Table T.1.

TABLE T.1

COMPLETE SEARCH STRINGS BY DATABASE (EXECUTED OCTOBER 2025)

Database	Search String
IEEE Xplore	("All Metadata":"artificial general intelligence" OR "All Metadata":"AGI" OR "All Metadata":"general purpose AI" OR "All Metadata":"human-level AI") AND ("All Metadata":"cyber-physical systems" OR "All Metadata":"social intelligence" OR "All Metadata":"cognitive architecture" OR "All Metadata":"embodied intelligence" OR "All Metadata":"multi-domain integration" OR "All Metadata":"cyber-physical-social")
ScienceDirect	TITLE-ABS-KEY (("artificial general intelligence" OR "AGI" OR "general artificial intelligence") AND ("cyber-physical" OR "social robot*" OR "theory of mind" OR "cognitive model" OR "embodied cognition" OR "cyber-physical-social systems" OR "cross-domain integration")) AND PUBYEAR > 2020
MDPI	("artificial general intelligence" OR "AGI") AND ("cyber-physical" OR "social intelligence" OR "cognitive architecture" OR "embodied AI" OR "multi-domain") in Abstract
SpringerLink	("artificial general intelligence" OR "AGI") AND ("cyber-physical systems" OR "social AI" OR "cognitive computing" OR "embodied intelligence" OR "cyber-physical-social") within Computer Science
ACM Digital Library	[Abstract: "artificial general intelligence" OR Abstract: "AGI" OR Abstract: "general purpose AI"] AND [Abstract: "cyber-physical" OR Abstract: "social intelligence" OR Abstract: "cognitive architecture" OR Abstract: "embodied intelligence" OR Abstract: "cross-domain"]

As highlighted in this Table T.2, this appendix presents the data extraction and categories.

TABLE T.2

DATA EXTRACTION CATEGORIES AND DESCRIPTIONS

Extraction Category	Description
Bibliographic Information	Authors, publication year, venue, publisher
Methodological Characteristics	Research design, evaluation methods, dataset characteristics
Architectural Specifications	System architecture, coordination mechanisms, integration approaches
CPST Dimension Coverage	Coverage of cyber, physical, social, and thinking dimensions
Cognitive Capabilities	Learning, reasoning, planning, and adaptation mechanisms
Evaluation Metrics	Performance measures, benchmarks, validation approaches
Implementation Challenges	Technical limitations, scalability issues, integration barriers
Research Gaps	Identified limitations and future research needs
Domain Applications	Specific application contexts and use cases